PROJECT PROPOSAL

Project Title

Smart Grocery Budget Planner

Why Smart?

"Smart" refers to the program's ability to automate calculations, making grocery expense tracking easier and more convenient. It refers to automated calculations, not artificial intelligence or advanced smart-device technology.

Intended Users

Grocery shoppers, individuals and families who want to manage their expenses, and students who want to practice budgeting. Based on our survey, 9 out of 12 respondants (75%) said that the proposed program would be helpful to them.

PROJECT DESCRIPTION

This Python program helps grocery shoppers, families, and students organize grocery lists and track their expenses. It allows users to enter item prices and quantities, calculate costs, and sort items by price. It uses lists, conditions, loops, and the math library to manage grocery information and perform calculations.

Problem Statement

People have different shopping habits and financial situations, and calculating grocery expenses can be time-consuming. The Smart Grocery Budget Planner helps grocery shoppers, families, and students organize grocery lists, calculate item costs based on price and quantity, and track total expenses. Users can manually enter and update prices, allowing them to reuse the program for different shopping trips without maintaining a permanent database of changing grocery prices.

Project Objectives

- Develop a python program that allows users to input grocery item names, prices, and quantities, automatically calculates the cost of each item, and determines the total grocery expenses.

- Ensure that the program accurately calculates individual item costs and the overall expenses, while successfully sorting grocery items according to their prices.
- Use basic Python programming concepts, such as variables, lists, loops, conditional statements, and sorting functions, to develop a functional grocery budget planner.
- Help users manage their grocery budgets more efficiently by automating calculations, organizing item prices, and making it easier to compare costs before purchasing.

Planned Features

- Add Item – Allows the user to enter an item name, price, and quantity.

- View Grocery List – Displays all added grocery items and their information.

- Compute Total Cost – Calculates the total cost of all grocery items.

- Sort Items by Price – Arranges items from the highest to the lowest price.

- Menu System – Allows users to choose different actions.

- Exit – Ends the program when the user is finished.

- Automatic Calculations – Calculates each item's total using price × quantity.

- Budget Tracking – Shows the user how much they are spending.

Planned Inputs and Outputs

Inputs

- Item Name – String

- Item Price – Float

- Quantity – Integer

- Menu Choice – Integer

Outputs

- Grocery item names

- Item prices

- Quantities

- Total price for each item

- Overall total cost
- Grocery items sorted from highest to lowest price
- Program messages such as item added successfully and program ended

LOGIC PLAN

```
START

IMPORT math

CREATE empty list item_names
CREATE empty list item_prices
CREATE empty list item_quantities

SET choice = 0

WHILE choice is not 5

    DISPLAY "========================================="
    DISPLAY "SMART GROCERY LIST & BUDGET"
    DISPLAY "1. Add Item"
    DISPLAY "2. View Grocery List"
    DISPLAY "3. Compute Total Cost"
    DISPLAY "4. Sort Items by Price"
    DISPLAY "5. Exit"
    DISPLAY "========================================="

    INPUT choice

    IF choice = 1 THEN

        INPUT item name
        INPUT item price
        INPUT quantity

        ADD item name to item_names
        ADD item price to item_prices
        ADD quantity to item_quantities

        DISPLAY "Item added successfully!"

    ELSE IF choice = 2 THEN
```

```
    DISPLAY "---------- GROCERY LIST ----------"

    FOR each item in the lists

        CALCULATE total_price = item_price × quantity

        DISPLAY item name
        DISPLAY item price
        DISPLAY quantity
        DISPLAY total_price

    END FOR

ELSE IF choice = 3 THEN

    SET overall_total = 0

    FOR each item in the lists

        CALCULATE total_price = item_price × quantity
        ADD total_price to overall_total

    END FOR

    DISPLAY "---------- TOTAL COST ----------"
    DISPLAY overall_total

ELSE IF choice = 4 THEN

    SORT item_names, item_prices, and item_quantities
    BASED ON item_prices FROM HIGHEST TO LOWEST

    DISPLAY "------ SORTED BY HIGHEST PRICE ------"

    FOR each item in the sorted lists

        DISPLAY item name
        DISPLAY item price

    END FOR

ELSE IF choice = 5 THEN
```

```
        DISPLAY "Thank you for using Smart Grocery List!"
        DISPLAY "Program ended."

    ELSE

        DISPLAY "Invalid choice. Please try again."

    END IF

END WHILE

END
```